Section 2 — System Architecture & C4 Diagrams

This section defines the complete system architecture of the Dark Cascade framework. Architecture is presented using the C4 Model (System Context → Container → Component) combined with the Purdue Model layer assignment and IEC 62443 security zone definitions. All component selections reference the Cerebro Pipeline Framework patterns validated in the GRT reference implementation (NVIDIA Jetson AGX Orin, Baumer VCXG, TensorRT, ZMQ MDP/0.2).

> **CORRECTION:** *Technology stack correction applied vs. URS v0.1: QuestDB is the hot-path per-cycle cache (replaces Redis reference). PostgreSQL is the relational store (Q-BOM, audit log, OEE aggregates). TimescaleDB (PostgreSQL extension) is the historian for 90-day time-series retention. Redis is NOT used in the Dark Cascade stack.*

> **NOTE:** *The Cerebro Pipeline Framework (ZMQ MDP/0.2, POSIX SHM, Cerebro Broker) is the inter-worker communication backbone. Direct worker-to-worker sockets are PROHIBITED. All messaging routes through the Cerebro Broker on ZMQ ROUTER :5555.*

2.1 Architecture Philosophy & Design Decisions

2.1.1 Core Principles

- Edge-first execution: SQI computation and vaenableAI inference execute on the Edge AI Gateway co-located with the machine. Zero network round-trips for production decisions. [URS-NFR-001, URS-NFR-002]
- Cerebro Broker as the single IPC fabric: All workers communicate exclusively through Cerebro MDP/0.2 (ZMQ ROUTER :5555). Priority queue: CRITICAL > HIGH > NORMAL > LOW. Circuit breakers per worker. Health check every 10s. [URS-NFR-008]
- Shared Memory for frame transport: Camera frames between Acquisition and Inference workers use POSIX SHM with seqlock and GSID continuity. ZMQ carries only control messages and results — never raw frames. [URS-NFR-002]
- Three-tier storage: QuestDB (hot cache, per-cycle writes at production rate, <5min window) → TimescaleDB (historian, 90-day hypertables, continuous aggregates) → PostgreSQL (relational — Q-BOM, audit, OEE). [URS-NFR-006, URS-VIS-008]
- Worker-per-concern: Each functional module is a discrete Cerebro worker with a unique service name and error code range. Failure isolation guaranteed — a crashed Q-BOM Worker does not affect the SQI pipeline. [URS-NFR-008]
- Python for business logic, C++ for latency-critical paths: Camera (Acquisition Worker, C++) and Inference (Inference Worker, C++ + TensorRT) require <100ms end-to-end. All other workers (SQI, Q-BOM, OEE, Alert, GSM, Historian) are Python 3.10. [CODING_CONVENTIONS §3/§4]

2.1.2 Architecture Decision Record (ADR)

ADR #	Decision	Rationale	Alternatives Rejected	Traces To
ADR-001	Cerebro MDP/0.2 over REST/gRPC for inter-worker IPC	Built-in priority queue, per-worker circuit breakers, health monitoring, reconnection with exponential backoff. Proven in GRT production. Zero external dependencies beyond libzmq.	REST microservices (no priority queue, no built-in health), gRPC (no priority, requires service mesh)	URS-NFR-008, URS-ALM-006
ADR-002	POSIX SHM + Seqlock for camera frame transport	Eliminates ZMQ copy overhead for ~60MB frames at 10 FPS+. Lock-free: one writer, N readers. GSID provides auditable frame continuity (gap rate target <0.1%). [FDS_TEMPLATE §5]	ZMQ PUB/SUB (copy-per-subscriber at 60MB = 600MB/s at 10fps), shared filesystem (latency, I/O bottleneck)	URS-VIS-001, URS-NFR-002
ADR-003	QuestDB as hot cache (NOT Redis)	Native time-series SQL (SAMPLE BY), ILP ingestion at millions of rows/s, per-cycle SQI and per-sachet DEOC writes without schema overhead. OEE Worker queries by time window natively.	Redis (no SQL, no time-series aggregation), InfluxDB OSS (license), TimescaleDB alone (write amplification)	URS-SQI-001, URS-OEE-001
ADR-004	TimescaleDB as historian	PostgreSQL extension — same psycopg3 connection pool as Q-BOM DB. Hypertables partition 90-day sachet-level history. Continuous aggregates for hourly/daily OEE rollups. No separate stack.	InfluxDB (separate stack, separate query language), QuestDB alone (no 90-day aggregation), OSIsoft PI (vendor lock-in, cost)	URS-VIS-008, URS-OEE-006
ADR-005	PostgreSQL for Q-BOM relational schema	ACID compliance for Q-BOM version history and immutable audit log. Row-level security for RBAC. JSON columns for flexible laminate thermal property storage.	SQLite (no concurrent writes), MongoDB (no ACID JOIN for version traceability)	URS-QBOM-001 through URS-QBOM-007
ADR-006	Multi-vendor PLC adapter	Dark Cascade supports AB (HUL Loop 3), Mitsubishi (HUL Loop 4), Omron (ULP	Separate PLC workers per vendor (multiplied	URS-ICD-001, GAP-G-N01

ADR #	Decision	Rationale	Alternatives Rejected	Traces To
	(pylogix/pyomron/MC Protocol)	Cavite) from a single PLC Worker. The Cerebro 'plc' service interface is identical regardless of PLC vendor — consumers always call the same REQUEST/REPLY schema.	maintenance), OPC-UA only (not universally available on all site PLCs)	
ADR-007	SQLite for Q-BOM offline cache on edge	Minimal footprint, zero external dependency. Single-record active_qbom_snapshot. SQI Worker reads from SQLite when network path to PostgreSQL is unavailable. [URS-NFR-006, TBD-008]	Redis (overkill for single-record cache, external process dependency), QuestDB (OLAP, not suitable for single-record key-value)	URS-NFR-006, URS-NFR-007

2.2 Purdue Model Layer Assignment (ISA-95)

Every Dark Cascade component is assigned to a Purdue Model level. Data flows upward (L0→L4); control flows downward (L4→L0). All inter-level communication traverses named conduits defined in §2.3.

Level	Name	Dark Cascade Components	Technology	Data Produced / Consumed	Direction
L0	**Field**	VFFS sealing jaws, filling piston, servo drives, Cimcon Vibit IoT sensors, Baumer VCXG cameras	*PLC I/O · GigE Vision · BLE/Wi-Fi (Vibit)*	Jaw torque, T_actual, dwell time, servo position, vibration RMS, camera frames	**→ L1**
L1	**Control**	PLC Worker ('plc') — multi-vendor adapterAcquisition Worker ('camera') — GigE Vision capture	*Python 3.10 (pylogix/pyomron/MC Protocol)C++17 + neoAPI + GigE Vision*	PLC tag values at poll rate (50ms)Raw frames via SHM-A	**→ L2**
L2	**Edge AI**	Cerebro Broker (ZMQ ROUTER :5555)SQI Engine ('sqi') · Inference ('inference') · Process ('process')Q-BOM ('qbom') · OEE ('oee') · Alert ('alert')GSM ('gsm') · Historian ('historian')QuestDB hot cache · SQLite offline cache	*C++17 (camera, inference, process)Python 3.10 (all other workers)POSIX SHM A + BQuestDB ILP :9009*	SQI per-cycle, DEOC per-sachet, alarms, OEE counters — all written to QuestDB	**→ L3 (async)**
L3	**Site Ops**	TimescaleDB historianPostgreSQL (Q-BOM + audit + OEE)WebSocket push server · REST/GraphQL APICockpit Dashboard server	*PostgreSQL 16 + TimescaleDBFastAPI + uvicornWebSocket Secure*	Historical SQI/DEOC trends (90 days)Q-BOM version history · Shift OEE reportsAlarm historian	**→ L4 (periodic)**
L4	**Enterprise**	AI4M Model Registry (ML weights)Cross-site BBOEE	*gRPC mTLS · HTTPS REST*	Updated TensorRT engines → L2Cross-site OEE benchmarks	**→ L2 (models**

Level	Name	Dark Cascade Components	Technology	Data Produced / Consumed	Direction
		dashboardGSM App (mobile/web)			)→ L3 (app)

2.3 IEC 62443 Security Zones & Conduits

Zone ID	Zone Name	Components	Purdue Levels	Trust Level	Conduits
Z-OT	**OT Zone**	VFFS PLCs, servo drives, filling piston, jaw actuators, laminate sensors	L0–L1	*Isolated OT subnet. Untrusted from IT perspective.*	→ Z-EDGE via C-01 (EtherNet/IP / MC Protocol / FINS, IP-whitelist-gated PLC writes)
Z-EDGE	**Edge AI Zone**	Cerebro Broker, all 11 workers, QuestDB, SQLite offline cache, Edge Gateway hardware	L1–L2	*High trust. Isolated VLAN. Production-critical.*	→ Z-OT via C-01 (PLC write, whitelist only)→ Z-SITE via C-02 (NATS JetStream TLS, REST HTTPS)
Z-SITE	**Site IT Zone**	TimescaleDB, PostgreSQL, WebSocket server, REST/GraphQL API, Cockpit Dashboard	L3	*Medium trust. Site LAN. IT-managed.*	← Z-EDGE via C-02 (TLS 1.3)→ Z-CLOUD via C-03 (HTTPS, mutual TLS)
Z-CLOUD	**Cloud / Central**	AI4M Model Registry, cross-site dashboard, GSM App	L4	*External. Zero implicit trust.*	← Z-SITE via C-03 (mutual TLS, JWT bearer)

> **NOTE:** *C-01 enforcement: (1) OT VLAN firewall restricts PLC write port access to Edge Gateway IP only. (2) PLC Worker application layer validates source and applies approved-register whitelist before every WRITE_TAGS command. [Traces to: URS-CL-002, URS-CL-003, URS-NFR-009, REG: IEC 62443-3-3 SR 3.1]*

2.4 C4 Level 1 — System Context Diagram (Mermaid)

Render in Mermaid Live (mermaid.live) or the Mermaid Chart tool. Audience: production managers, site directors, client sponsors.

```mermaid
C4Context
  title Dark Cascade Framework — System Context

  Person(cockpit_op, "Cockpit Operator",
    "Production supervisor — monitors BBOEE, applies AI prescriptions")
  Person(field_op, "Field Operator",
    "Line technician — receives work instructions on tablet, clears alarms")
  Person(supervisor, "Quality Supervisor",
    "Activates Q-BOMs, sets correction bounds, reviews shift reports")

  System(dc, "Dark Cascade Framework",
    "EnableAI + vaenableAI — autonomous VFFS quality intelligence")

  System_Ext(vffs, "VFFS Machine(s)",
    "S26 Shubham (ULP) / 12x VFFS CLS-1 (HUL) — PLCs, cameras, sensors")
  System_Ext(model_reg, "AI4M Model Registry",
    "Cloud — TensorRT engine versions, SQI physics model updates")
  System_Ext(gsm_app, "GSM App",
    "Mobile/web — laminate QR scan, parameter calculator")
  System_Ext(cockpit, "Cockpit Dashboard",
    "Web — BBOEE, OEE, SQI trend, alarm stream, prescription apply")
  System_Ext(tablet, "Field Operator Tablet",
    "Android/iOS — bounding-box alerts, work instructions, Clear-OK")

  Rel(cockpit_op, cockpit, "Views KPIs and applies prescriptions", "HTTPS/WSS")
  Rel(field_op, tablet, "Acknowledges alarms, clears faults", "WebSocket push")
  Rel(supervisor, gsm_app, "Scans laminate QR, activates Q-BOM", "HTTPS REST")
  Rel(dc, vffs, "Reads PLC tags; writes corrective params (whitelist-gated)",
    "EtherNet/IP · MC Protocol · FINS/TCP")
  Rel(vffs, dc, "Camera frames; PLC telemetry; vibration RMS",
    "GigE Vision · PLC poll · BLE/Wi-Fi")
  Rel(dc, cockpit, "Pushes KPI updates, alarms, shift reports", "WSS · REST")
  Rel(dc, tablet, "Pushes work instructions with defect images", "WSS")
  Rel(dc, gsm_app, "Q-BOM CRUD API; parameter compute responses", "HTTPS REST")
  Rel(model_reg, dc, "Model weight updates (TensorRT engines)", "gRPC mTLS")
```

2.5 C4 Level 2 — Container Diagram — Edge AI Gateway (Mermaid)

Audience: architects, DevOps, site IT. Shows all containers on the RHEL 10 / Podman Quadlet edge node.

```mermaid
C4Container
  title Dark Cascade — Edge AI Gateway Containers
```

```
  Container_Boundary(edge, "Edge AI Gateway [RHEL 10 / Podman Quadlet]") {

    Container(cerebro, "Cerebro Broker",
      "C++17 / ZMQ ROUTER :5555",
      "MDP/0.2 — priority queuing, circuit breakers, health monitoring")

    Container(plc_w,  "PLC Worker",       "Python / pyzmq",
      "Multi-vendor adapter: pylogix(AB), pyomron(Omron), MC Protocol(Mitsubishi). Service: 'plc'")
    Container(cam_w,  "Acquisition Worker","C++17 / neoAPI",
      "GigE Vision capture, HW trigger, SHM-A writer, GSID. Service: 'camera'")
    Container(infer_w,"Inference Worker",  "C++17 / TensorRT 8",
      "SegFormer DEOC + YOLO corners, SHM-A reader, SHM-B writer. Service: 'inference'")
    Container(proc_w, "Process Worker",    "C++17 / pyzmq",
      "CRQS classification, DEOC grading, S3 image upload. Service: 'process'")
    Container(sqi_w,  "SQI Engine Worker", "Python 3.10",
      "EnableAI physics model, failure mode, alarm trigger. Service: 'sqi'")
    Container(qbom_w, "Q-BOM Worker",      "Python 3.10 / FastAPI",
      "Q-BOM CRUD, lifecycle state machine, REST API. Service: 'qbom'")
    Container(alert_w,"Alert Worker",      "Python 3.10 / websockets",
      "Dual-stream routing, WSS push, escalation timer. Service: 'alert'")
    Container(oee_w,  "OEE Worker",        "Python 3.10",
      "OEE/BBOEE, loss categorization, waste formula. Service: 'oee'")
    Container(gsm_w,  "GSM Worker",        "Python 3.10 / scikit-learn",
      "QR intake, regression compute, Draft Q-BOM, PLC push. Service: 'gsm'")
    Container(hist_w, "Historian Worker",  "Python 3.10",
      "QuestDB to TimescaleDB async sync, retention enforcement. Service: 'historian'")

    ContainerDb(questdb, "QuestDB", "Hot cache",
      "ILP :9009 — SQI/DEOC writes at production rate. SQL :8812 — OEE queries")
    ContainerDb(shm_a, "SHM-A", "POSIX SHM", "~60MB/slot — camera frames, seqlock, GSID")
    ContainerDb(shm_b, "SHM-B", "POSIX SHM", "~8MB/slot — inference results, bbox, confidence")
    ContainerDb(sqlite, "Q-BOM Offline Cache", "SQLite",
      "Active Q-BOM snapshot — SQI Worker reads on offline fallback (8h+)")
  }

  ContainerDb(pg, "PostgreSQL + TimescaleDB", "Site Server",
    "Q-BOM schema + audit_log (PostgreSQL) + SQI/DEOC historian hypertables (TimescaleDB)")
  ContainerDb(s3, "Object Storage", "S3-compatible",
    "Defect images — Red + Amber events, 90-day retention")

  Rel(cam_w,  shm_a,  "Frame write",   "SHM seqlock")
  Rel(infer_w,shm_a,  "Frame read",    "SHM seqlock")
  Rel(infer_w,shm_b,  "Result write",  "SHM seqlock")
  Rel(proc_w, shm_b,  "Result read",   "SHM seqlock")
  Rel(proc_w, questdb,"DEOC write",    "ILP :9009")
  Rel(sqi_w,  questdb,"SQI write",     "ILP :9009")
  Rel(oee_w,  questdb,"OEE read",      "SQL :8812")
  Rel(hist_w, questdb,"Sync read",     "SQL :8812")
  Rel(hist_w, pg,     "Historian write","psycopg3")
  Rel(qbom_w, pg,     "Q-BOM CRUD",   "psycopg3")
  Rel(sqi_w,  sqlite, "Offline Q-BOM","SQLite")
  Rel(proc_w, s3,     "Defect images", "S3 API HTTPS")
```

2.6 C4 Level 3 — Component: Cerebro Broker & Worker Topology (Mermaid)

Audience: developers, systems engineers. Shows how each worker connects to the Cerebro Broker and the data routing model.

```mermaid
flowchart LR
  subgraph CLI["Clients (DEALER sockets)"]
    C_GSM["GSM App\n[HTTP → qbom API]"]
    C_DASH["Cockpit Dashboard\n[WSS → alert]"]
    C_SQI["SQI Worker\n(reads plc tags)"]
    C_PROC["Process Worker\n(reads inference)"]
  end

  subgraph BRK["Cerebro Broker — ZMQ ROUTER :5555"]
    PQ["Priority Queue\nCRITICAL > HIGH > NORMAL > LOW"]
    CB_BRK["Per-Worker Circuit Breakers\n(5 failures/30s → OPEN)"]
    HM["HEALTH_CHECK every 10s\n(workers MUST respond within 2s)"]
    DISP["Dispatch Algorithm\nscore = health×10 + locality\n+ GPU_match - queue_depth×0.1"]
    PQ --> DISP
    CB_BRK --> DISP
    HM --> CB_BRK
  end

  subgraph WRK["Workers — DEALER sockets (READY on startup, HEARTBEAT every 2s)"]
    W1["PLC Worker  'plc'\n4xxx errors"]
    W2["Acquisition  'camera'\n2xxx errors"]
    W3["Inference  'inference'\n3xxx errors"]
    W4["Process  'process'\n6xxx errors"]
    W5["SQI Engine  'sqi'\n7xxx errors"]
    W6["Q-BOM  'qbom'\n8xxx errors"]
    W7["Alert  'alert'\n9xxx errors"]
    W8["OEE  'oee'\n10xxx errors"]
    W9["GSM  'gsm'\n11xxx errors"]
    W10["Historian  'historian'\n12xxx errors"]
  end

  CLI -->|"REQUEST frames"| BRK
  BRK -->|"REQUEST frames"| WRK
  WRK -->|"REPLY / HEARTBEAT / DISCONNECT"| BRK
  BRK -->|"REPLY to client"| CLI
```

2.7 Worker Inventory — Cerebro Service Registry

Authoritative registry of all 11 Cerebro workers in Dark Cascade. Startup order is enforced by Podman Quadlet dependency declarations. HEALTH_CHECK response is MANDATORY for all workers per RESILIENCE.md §6.

#	Worker Name	Service	Language	Error Range	HW Dep.	Startup After	Responsibilities
1	**Cerebro Broker**	*(internal)*	C++17	1xxx	None	—	ZMQ ROUTER :5555 · Priority queuing · Per-worker circuit breakers (5 fail/30s) · HEALTH_CHECK every 10s · Worker expiry 10s
2	**PLC Worker**	*'plc'*	Python 3.10	4xxx	EtherNet/IP / MC Protocol / FINS network	*Broker*	Multi-vendor tag read/write (pylogix/pyomron/MC Protocol) · Circuit breaker on PLC connection · HEARTBEAT 2s · HEALTH_CHECK response MANDATORY · DISCONNECT on shutdown
3	**Acquisition Worker**	*'camera'*	C++17	2xxx	Baumer VCXG GigE Vision · 10Gbps SFP+ fiber	*Broker · SHM-A allocated*	HW trigger (PLC digital out) · Camera parameter apply from Q-BOM HW_Config · SHM-A frame write · GSID increment · Circuit breaker on camera hardware
4	**Inference Worker**	*'inference'*	C++17 + TensorRT	3xxx	NVIDIA GPU (SM 8.7 Orin / SM 7.2 Xavier)	*Acquisition Worker · SHM-A*	SHM-A read → GPU preprocess → SegFormer DEOC (512×512) + YOLO corners → SHM-B write · Model load from Q-BOM active SKU · TensorRT FP16
5	**Process Worker**	*'process'*	C++17	6xxx	SHM-B · QuestDB :9009 · S3 API	*Inference Worker · SHM-B · QuestDB*	CRQS classification · DEOC grade assignment · Defect image async upload to S3 · GSID continuity check · QuestDB ILP write (deoc_results)
6	**SQI Engine Worker**	*'sqi'*	Python 3.10	7xxx	QuestDB :9009 · SQLite cache	*Broker · PLC Worker · QuestDB*	Per-cycle SQI physics computation f(T,P,t,λ,C,α,A,d,k,T_amb) · Green/Amber/Red grading · Failure mode classification · Alarm trigger via Alert Worker · PLC write orchestration
7	**Q-BOM Worker**	*'qbom'*	Python 3.10	8xxx	PostgreSQL · SQLite cache sync	*Broker · PostgreSQL*	Q-BOM CRUD (4 entities) · Lifecycle state machine (Draft→Pending→Active→Archived) · FastAPI REST northbound · Offline cache sync after every change
8	**Alert Worker**	*'alert'*	Python 3.10	9xxx	WebSocket server (site LAN)	*Broker · SQI · Process Workers*	Dual-stream routing (Cockpit + Field) · WSS push ≤2s · 3 retries → Cockpit escalation · Escalation timer (10min default) · Clear-OK logging · Alarm historian write
9	**OEE Worker**	*'oee'*	Python 3.10	10xxx	QuestDB :8812 · PostgreSQL	*Broker · PLC Worker · QuestDB*	OEE = Avail × Perf × Quality (ISA-95) · BBOEE rolling 30-day max · Strategic/Non-Strategic

#	Worker Name	Service	Language	Error Range	HW Dep.	Startup After	Responsibilities
							categorization · Wasted laminate formula · Shift report trigger
10	**GSM Worker**	***'gsm'***	Python 3.10	**11xxx**	Q-BOM Worker · PLC Worker	*Broker · Q-BOM Worker*	QR scan intake · scikit-learn polynomial regression compute (≤120s) · Draft Q-BOM creation · Changeover PLC push orchestration via PLC Worker
11	**Historian Worker**	***'historian'***	Python 3.10	**12xxx**	QuestDB :8812 · TimescaleDB	*Broker · QuestDB · TimescaleDB*	Async QuestDB→TimescaleDB batch sync (5-min intervals) · 30-day downsampling · 90-day retention enforcement · Continuous aggregate refresh

2.8 Full Technology Stack

2.8.1 Platform, Runtime & Frameworks

Category	Component	Version	Role in Dark Cascade	URS/ADR Ref
OS	RHEL 10	GA	Edge Gateway — SELinux enforced, FIPS-compliant, Podman-native	URS-NFR-009
Containerisation	Podman + Quadlet	Podman 5.x	Rootless container orchestration; Quadlet units for dependency ordering	ADR-007
IPC Messaging	ZeroMQ (libzmq)	4.3.5	Cerebro broker ROUTER :5555 + all worker DEALER sockets	ADR-001
Broker Framework	Cerebro (internal)	v1.0	MDP/0.2, priority queuing, circuit breakers, health monitoring, reconnection backoff	ADR-001
Frame Transport	POSIX SHM + Seqlock	Kernel 5.14+	SHM-A (~60MB/slot) and SHM-B (~8MB/slot) — lock-free, GSID continuity	ADR-002
C++ Standard	C++17 (gcc 13)	C++17	Camera acquisition, inference, process workers	CODING_CONVENTIONS §3
Python Standard	Python 3.10 (CPython)	3.10	All business logic workers (SQI, Q-BOM, OEE, Alert, GSM, Historian)	CODING_CONVENTIONS §4
JSON Library (C++)	nlohmann/json	v3.11.2	CerebroMessage/CerebroResponse serialization in all C++ workers	CODING_CONVENTIONS §5
ZMQ Python binding	pyzmq	>=25.0	All Python worker DEALER sockets + reconnection backoff	RESILIENCE.md §1.2
API Framework	FastAPI + uvicorn	FastAPI 0.110	Q-BOM Worker northbound REST API (HTTPS :8443)	ADR-005

2.8.2 AI / ML Stack

Category	Component	Version	Role	URS Ref
GPU Inference	TensorRT	8.x (NVIDIA)	vaenableAI inference — SegFormer (DEOC patches, FP16) + YOLO (corners). GPU SM 8.7 Orin / SM 7.2 Xavier.	URS-VIS-001, URS-VIS-002
Camera SDK	Baumer neoAPI	v1.5.0	Acquisition Worker — GigE Vision, hardware trigger, exposure, strobe control from Q-BOM HW_Config	URS-VIS-004
Vision Framework	NVIDIA DeepStream	6.x (optional)	Multi-camera pipeline orchestration at HUL Haridwar (12 machines)	HUL site config
SQI Physics Engine	EnableAI SQI v1.0	AI4M proprietary	Thermal energy transfer f(T_actual, P_torque, t_dwell, λ, C, α, A, d, k, T_amb) — Python implementation	URS-SQI-003
Parameter Regression	scikit-learn	>=1.3	GSM Worker — polynomial regression for Golden Parameter computation (seal_temp, stroke, CPM)	URS-GSM-002
Model Registry	AI4M Model Registry	gRPC/HTTPS	TensorRT engine versioning; Inference Worker downloads on UPDATE_MODEL command (mTLS)	ICD-011

2.8.3 Data Storage Stack

Storage Role	Component	Version	Schema / Use	URS/ADR Ref
Hot Cache (L2)	QuestDB	7.x	sqi_cycles · deoc_results · oee_counters ·	ADR-003, URS-SQI-001

Storage Role	Component	Version	Schema / Use	URS/ADR Ref
			servo_metrics — all at production rate. ILP :9009 write, SQL :8812 read. <5min rolling window.	
Relational (L3)	PostgreSQL	16	qbom_schema (4 entities) · audit_log (immutable append-only) · oee_aggregates · user_roles — ACID, row-level security.	ADR-005, URS-QBOM-001
Historian (L3)	TimescaleDB	2.x (PG extension)	sqi_history · deoc_history · oee_daily — hypertables by day. Continuous aggregates at 1h/24h. 90-day retention.	ADR-004, URS-VIS-008
Offline Cache (L2)	SQLite	3.x	active_qbom_snapshot (single row) — Q-BOM Worker syncs after every Active promotion. SQI Worker reads on network failure.	ADR-007, URS-NFR-006
Object Storage (L3/L4)	S3-compatible (MinIO / Azure Blob)	API v4	defect_images — {sku_id, timestamp, machine_id, deoc_grade, defect_class_list, confidence_score}. 90-day retention policy enforced by bucket lifecycle rule.	URS-VIS-008

2.9 Shared Memory Configuration

Segment	SHM Name	Producer	Consumer	Slot Count	Frame Size (typ.)	Synchronisation & GSID Contract
SHM-A	/shm_dc_frames_a	Acquisition Worker ('camera')	Inference Worker ('inference')	4	~60MB (5250×2650×3 RGB)	Seqlock (lock-free: 1 writer, N readers). Producer increments GSID per frame on commit. Consumer checks GSID delta: delta > 1 = frame drop event logged to QuestDB gsid_gaps table. GSID gap rate target: <0.1% per FDS_TEMPLATE Appendix A.
SHM-B	/shm_dc_results_b	Inference Worker ('inference')	Process Worker ('process')	4	~8MB (seg mask + bbox JSON)	Same seqlock + GSID pattern. Result payload: {segmentation_mask_bytes, bboxes_json, class_scores_json, inference_time_ms, model_version_hash, source_gsid}.

> **NOTE:** *SHM-A frame size for HUL Haridwar (multi-camera, sensor TBD) will be recalculated at hardware specification time. Slot count must satisfy: slot_count >= ceil(pipeline_latency_ms / frame_period_ms) + 1 to prevent producer overrun. [FDS_TEMPLATE §5.2]*

2.10 Error Code Allocation — Dark Cascade Extensions to Cerebro Protocol

Error codes are STRINGS per CODING_CONVENTIONS §2 (e.g. "7003" not 7003). Dark Cascade extends Cerebro's 1xxx–6xxx ranges with the following new service allocations:

Range	Service Name	Worker	Status	Initial Error Codes Defined
1xxx	system	Cerebro Broker	Existing	1000 UNKNOWN_ERROR · 1001 SERVICE_UNAVAILABLE · 1002 TIMEOUT · 1003 INVALID_MESSAGE · 1004 QUEUE_FULL (see PROTOCOL.md §6.1)
2xxx	camera	Acquisition Worker	Existing	2000–2006 — see PROTOCOL.md §6.2

Range	Service Name	Worker	Status	Initial Error Codes Defined
3xxx	inference	Inference Worker	Existing	3000–3005 — see PROTOCOL.md §6.3
4xxx	plc	PLC Worker	Existing	4000–4006 — see PROTOCOL.md §6.4
5xxx	database	(legacy cerebrum)	Existing	5000–5005 — see PROTOCOL.md §6.5
6xxx	process	Process Worker	Existing	6000–6016 — see PROTOCOL.md §6.6
7xxx	sqi	SQI Engine Worker	NEW — Dark Cascade	7000 SQI_ERROR · 7001 PLC_DATA_STALE · 7002 QBOM_NOT_LOADED · 7003 THRESHOLD_NOT_CALIBRATED · 7004 PHYSICS_COMPUTE_FAILED · 7005 SQI_OUT_OF_RANGE · 7006 OFFLINE_MODE_ACTIVE
8xxx	qbom	Q-BOM Worker	NEW — Dark Cascade	8000 QBOM_ERROR · 8001 SKU_NOT_FOUND · 8002 QBOM_NOT_ACTIVE · 8003 LIFECYCLE_TRANSITION_DENIED · 8004 RBAC_PERMISSION_DENIED · 8005 VERSION_CONFLICT · 8006 LAMINATE_QR_NOT_FOUND · 8007 DUAL_SIGNOFF_REQUIRED
9xxx	alert	Alert Worker	NEW — Dark Cascade	9000 ALERT_ERROR · 9001 WEBSOCKET_DELIVERY_FAILED · 9002 ESCALATION_TIMEOUT · 9003 ACK_NOT_FOUND · 9004 STREAM_ROUTING_FAILED · 9005 ALARM_NOT_FOUND
10xxx	oee	OEE Worker	NEW — Dark Cascade	10000 OEE_ERROR · 10001 STATUS_CODE_UNMAPPED · 10002 PLC_COUNTER_STALE · 10003 BBOEE_INSUFFICIENT_DATA · 10004 SHIFT_REPORT_FAILED
11xxx	gsm	GSM Worker	NEW — Dark Cascade	11000 GSM_ERROR · 11001 LAMINATE_PROPS_INCOMPLETE · 11002 REGRESSION_MODEL_NOT_LOADED · 11003 COMPUTE_TIMEOUT · 11004 PLC_PUSH_FAILED · 11005 DRAFT_QBOM_CREATION_FAILED
12xxx	historian	Historian Worker	NEW — Dark Cascade	12000 HISTORIAN_ERROR · 12001 QUESTDB_READ_FAILED · 12002 TIMESCALEDB_WRITE_FAILED · 12003

Range	Service Name	Worker	Status	Initial Error Codes Defined
				SYNC_BATCH_TIMEOUT · 12004 RETENTION_ENFORCEMENT_FAILED

2.11 Key Sequence Diagrams (Mermaid)

2.11.1 Per-Cycle SQI Computation — Full Data Flow (≤100ms budget)

```mermaid
sequenceDiagram
  participant PLC  as VFFS PLC
  participant PLCw as PLC Worker  'plc'
  participant BRK  as Cerebro Broker
  participant SQIw as SQI Engine  'sqi'
  participant QDB  as QuestDB ILP :9009
  participant ALT  as Alert Worker  'alert'
  participant WS   as WebSocket Server

  PLC->>PLCw: Jaw close — trigger poll (every 50ms)
  PLCw->>PLCw: Read T_actual, P_torque, t_dwell, T_ambient via pylogix/pyomron/MC
  PLCw->>BRK:  REPLY 'plc' GET_CYCLE_TAGS (tag_payload)
  BRK->>SQIw:  REQUEST — cycle tag payload
  SQIw->>SQIw: Load laminate props from SQLite offline cache (<5ms)
  SQIw->>SQIw: SQI = f(T, P, t, λ, C, α, A, d, k, T_amb) [≤50ms physics compute]
  SQIw->>SQIw: Grade vs Q-BOM thresholds → Green / Amber / Red
  SQIw->>SQIw: Classify failure_mode if non-Green
  SQIw->>QDB:  ILP write: sqi_cycles(ts, sqi, grade, failure_mode, cycle_id, sku_id)
  alt consecutive_red >= 2  [Major Alarm]
    SQIw->>BRK:  REQUEST 'alert' MAJOR_ALARM {sqi, failure_mode, prescription} [priority: CRITICAL]
    BRK->>ALT:   Route Major Alarm
    ALT->>WS:    WSS push to Cockpit + Field tablet (≤2s delivery SLA)
    ALT->>BRK:   REQUEST 'sqi' TRIGGER_CLOSED_LOOP (if auto_correction_enabled)
  else consecutive_amber >= 3  [Minor Alarm]
    SQIw->>BRK:  REQUEST 'alert' MINOR_ALARM [priority: HIGH]
    BRK->>ALT:   Route → Field tablet only
  end
  Note right of SQIw: Budget: PLC poll ≤50ms + compute ≤50ms = ≤100ms total
```

2.11.2 vaenableAI Per-Sachet Inspection Flow (≤200ms budget)

```mermaid
sequenceDiagram
  participant PLC  as VFFS PLC
  participant CAMw as Acquisition Worker  'camera'
  participant SHMA as SHM-A
  participant INFw as Inference Worker  'inference'
  participant SHMB as SHM-B
  participant PROw as Process Worker  'process'
  participant QDB  as QuestDB ILP :9009
  participant S3   as Object Storage
  participant BRK  as Cerebro Broker
  participant ALT  as Alert Worker  'alert'

  PLC->>CAMw:  H-Sealer digital output pulse (hardware trigger)
  CAMw->>CAMw: Apply trigger_delay_ms from Q-BOM HW_Config (default 5ms)
  CAMw->>CAMw: Capture frame at Q-BOM exposure_us setting
```

```
  CAMw->>SHMA: Write frame + GSID (seqlock commit, ~60MB)
  INFw->>SHMA: Detect new GSID via seqlock poll
  INFw->>INFw: GPU preprocess: resize to 512×512 patches (batched)
  INFw->>INFw: TensorRT: SegFormer DEOC + YOLO corners (~25ms FP16)
  INFw->>SHMB: Write {seg_mask, bboxes, class_scores, inference_ms, model_hash, source_gsid}
  PROw->>SHMB: Detect new GSID via seqlock poll
  PROw->>PROw: CRQS classify — apply Q-BOM thresholds per defect class
  PROw->>PROw: DEOC grade: Green / Amber / Red
  PROw->>PROw: Check GSID continuity (gap > 1 → log gsid_gap event)
  PROw->>QDB:  ILP write: deoc_results(ts, deoc_grade, defect_classes, confidence, sachet_id, sku_id)
  alt deoc_grade = Amber OR Red
    PROw->>S3:   Async upload defect image {sku_id, ts, machine_id, grade, classes}
    PROw->>BRK:  REQUEST 'alert' DEOC_ALERT {grade, defect_classes, image_url} [priority: HIGH]
    BRK->>ALT:   Route alarm to appropriate stream
  end
  Note right of INFw: Inference: ~25ms | Classify: ~5ms | Total budget: ≤200ms
```

2.11.3 Closed-Loop PLC Correction Flow

```mermaid
sequenceDiagram
  participant SQIw as SQI Engine  'sqi'
  participant BRK  as Cerebro Broker
  participant QBOMw as Q-BOM Worker  'qbom'
  participant PLCw as PLC Worker  'plc'
  participant PLC  as VFFS PLC
  participant ALT  as Alert Worker  'alert'
  participant PG   as PostgreSQL  audit_log

  SQIw->>BRK:   REQUEST 'qbom' GET_ACTIVE_BOUNDS {sku_id} [priority: HIGH]
  BRK->>QBOMw:  Route
  QBOMw->>BRK:  REPLY {max_correction_bound_pct, baseline_seal_temp, auto_correction_enabled}
  BRK->>SQIw:   Deliver bounds
  SQIw->>SQIw:  Compute required_correction (e.g. +2°C on SEAL_TEMP_SP)
  SQIw->>SQIw:  Bounds check: |correction_pct| <= max_correction_bound_pct
  alt Within bounds AND auto_correction_enabled = TRUE
    SQIw->>BRK:  REQUEST 'plc' WRITE_TAGS {register: SEAL_TEMP_SP, value: new_val} [priority: HIGH]
    BRK->>PLCw:  Route (broker circuit breaker checked)
    PLCw->>PLCw: Validate register against approved whitelist
    PLCw->>PLC:  EtherNet/IP or MC Protocol write to register
    PLC-->>PLCw: Write confirmation
    PLCw->>BRK:  REPLY {success: true, written_value}
    PLCw->>BRK:  REQUEST 'plc' READ_TAGS {register: SEAL_TEMP_SP}  [readback, ≤500ms]
    PLCw-->>SQIw: Readback value delivered
    SQIw->>SQIw:  Verify readback within 2% of commanded value
    SQIw->>PG:    INSERT audit_log {ts, machine_id, register, prior_val, written_val,
                  correction_pct, triggering_sqi, failure_mode, operator_override=false}
                  [IMMUTABLE — no UPDATE or DELETE permitted]
  else Exceeds bounds OR auto_correction_enabled = FALSE
    SQIw->>BRK:  REQUEST 'alert' RED_ALERT {reason: BOUNDS_EXCEEDED, prescription: advice}
    BRK->>ALT:   Route to Cockpit (advisory prescription + one-click apply)
  end
```

2.12 Port Map — Dark Cascade Edge AI Gateway

Port	Protocol	Direction	Zone	Used By	Purpose
:5555	TCP (ZMQ)	Internal	Z-EDGE	Cerebro Broker	ZMQ ROUTER — all 11 workers connect as DEALER sockets. Cerebro MDP/0.2.
:9009	TCP (ILP)	Edge→QuestDB	Z-EDGE	SQI Worker, Process Worker	QuestDB ILP ingestion — sqi_cycles, deoc_results, oee_counters at production rate.
:8812	TCP (SQL)	Edge→QuestDB	Z-EDGE	OEE Worker, Historian Worker	QuestDB SQL interface — SAMPLE BY time-window queries.
:5432	TCP	Edge→Site	Z-EDGE→Z-SITE	Q-BOM Worker, Historian Worker	PostgreSQL 16 + TimescaleDB — Q-BOM schema, audit_log, sqi_history hypertable.
:4222	TCP (NATS TLS)	Edge→Site	Z-EDGE→Z-SITE	Historian Worker	NATS JetStream TLS — async telemetry streaming edge→site server for batch sync.
:8443	TCP (HTTPS)	Site→Clients	Z-SITE	Q-BOM Worker REST API	FastAPI REST — Q-BOM CRUD, GSM App, Cockpit Dashboard. JWT auth.
:8765	TCP (WSS)	Site→Clients	Z-SITE	Alert Worker WebSocket	WebSocket Secure — real-time alarm push to Cockpit Dashboard and Field tablet. ≤2s SLA.
:9000	TCP (gRPC mTLS)	Cloud→Edge	Z-CLOUD→Z-EDGE	Inference Worker	AI4M Model Registry — TensorRT weight downloads on UPDATE_MODEL command.
:44818	TCP	Edge→OT	Z-EDGE→Z-OT (C-01)	PLC Worker (pylogix)	EtherNet/IP CIP — Allen Bradley PLC tag read + approved-register write (HUL Loop 3).
:5007	TCP	Edge→OT	Z-EDGE→Z-OT (C-01)	PLC Worker (MC Protocol)	Mitsubishi MC Protocol — PLC tag read + approved-register write (HUL Loop 4).

Port	Protocol	Direction	Zone	Used By	Purpose
:9600	UDP	Edge→OT	Z-EDGE →Z-OT (C-01)	PLC Worker (pyomron)	Omron FINS/TCP — NJ501-1500 tag read + approved-register write (ULP Cavite).
GigE	10Gbps SFP+ fiber	OT→Edge (C-01-V)	Z-OT→Z-EDGE	Acquisition Worker	Dedicated fiber — Baumer VCXG uncompressed frames. Hardware trigger (PLC digital out). No write path.

2.13 Section 2 Approval — Gate G-01

By signing below, the reviewer confirms that architecture decisions, C4 diagrams, technology stack, security zones, and worker registry correctly represent the intended Dark Cascade system design. Gate G-01 approval permits Section 3 (Q-BOM Functional Specification) generation.

Role	Name	Signature	Date / Status
AI4M Project Lead	Suraj Rathod	_____	PENDING
AI4M Architect	[TBD]	_____	PENDING
Site IT/OT Lead (ULP Cavite)	[TBD]	_____	PENDING
Site IT/OT Lead (HUL Haridwar)	[TBD]	_____	PENDING

2.14 Version History — Section 2

Version	Date	Author	Status	Changes
v0.1	2026-04-04	AI4M Technology	**DRAFT**	Initial issue. Cerebro MDP/0.2 framework adopted as IPC backbone. 7 ADRs documented. Purdue L0-L4 placement. IEC 62443 zones (Z-OT, Z-EDGE, Z-SITE, Z-CLOUD). 11-worker registry with error ranges 7xxx–12xxx. Storage stack: QuestDB hot cache, PostgreSQL Q-BOM/audit, TimescaleDB historian, SQLite offline cache. 3 sequence diagrams (SQI, DEOC, closed-loop). Port map 12 entries. Technology corrections applied from Cerebro framework docs (PROTOCOL.md, RESILIENCE.md, CODING_CONVENTIONS.md, FDS_TEMPLATE.md).